

Tutorial Serie 2

Objectives: to master the simple conditional instruction, alternative, Multiple choice (according to)

Exercise 1

Write an algorithm that calculates and displays the absolute value of the difference between two real numbers entered on the keyboard.

Exercise 2

Let A, B and C be three points in the plane defined by their coordinates $A(x_A, y_A)$, $B(x_B, y_B)$, $C(x_C, y_C)$.
Write an algorithm to determine and display the nature of the ABC triangle.
An ABC triangle is either isosceles (has two equal sides), equilateral (has three equal sides) or Scalene (has no equal sides).

Exercise 3

Write a C program that asks the user for a child's age (in years). Then he informs him of his sporting category (using the **switch** instruction):

- "Chicken" : from 6 to 7 years
- "Pupil" : from 8 to 9 years
- "Minimum" : from 10 to 11 years
- "Younger" : from 12 to 14 years
- "Junior" : from 15 to 17 years

Supplementary Exercises

Exercise 1

Write an algorithm that allows to sort in ascending order two integers (A, B) read from the keyboard ($A \leq B$).

Exercise 2

Write an algorithm that sorts in descending order three integers (A , B , and C) read from the keyboard. A contains the highest value, and C contains the smallest ($A \geq B \geq C$).

Exercise 3

In a supermarket checkout, we receive a discount on the total amount of our purchases when they exceed 1000 Algerian Dinars (DA).

Write an algorithm that, after reading the amount of our purchases, calculates and displays the amount to pay. The discount is:

- 5% for a purchase less than or equal to 3000 DA,
- 7% for a purchase greater than 3000 and less than or equal to 15000 DA
- 15% for a purchase exceeding 15000 DA.

Exercise 4

Write an algorithm that reads a real number and:

- *Checks and displays whether the number is positive, negative, or zero.*
- *Checks and displays whether the number is even or odd.*
- *Displays its quarter, its third, and its half.*

Exercise 5

Write an algorithm that reads two real numbers (X and Y) and:

- *Calculates and displays the sum of X and Y if they are strictly positive.*
- *Calculates and displays the product of X and Y if they are not strictly positive.*
- *Calculates and displays the difference between these two numbers.*

Exercise 6

*Write an algorithm that displays the type of a letter (vowel or consonant) entered by the user using the **switch** statement.*

Exercise 7

Write an algorithm that asks the user to enter a character, then displays whether it is an uppercase or lowercase letter. If it is not a letter, it returns an error message

Exercise 8

Let X be a strictly positive integer.

- *Write an algorithm that reads the value of X and checks if X is divisible by both 5 and 7 without using logical operators.*
- *Write an algorithm that reads the value of X and checks if X is divisible by both 5 and 7 using logical operators.*
- *Write an algorithm that reads the value of X and checks if X is divisible by either 5 or 7 without using logical operators.*
- *Write an algorithm that reads the value of X and checks if X is divisible by either 5 or 7 using logical operators.*

Exercise 9

Write an algorithm that reads two real numbers m and n from the user and informs whether their product (without calculating the product) is negative, positive, or zero.

Exercise 10

Write an algorithm that asks the user for the width and length of a rectangle (in meters), then calculates and displays the perimeter and the area of the rectangle.

Exercise 11

Write an algorithm that calculates and displays the volume of a sphere.

Exercise 12

Write an algorithm that allows the user to enter three real numbers, then based on a choice from a menu displayed on the screen, calculates and displays the sum, product, average, finds the minimum, or finds the maximum of the three numbers.

The menu should appear on the screen as follows:

```

----- MENU -----
1 ----- Sum -----
2 ----- Product -----
3 ----- Average -----
4 ----- Minimum -----
5 ----- Maximum -----
Enter your choice:

```

Exercise 13

Let the following three algorithms, proposed by a computer science amateur, aim to:

- *Max_2_integers*: Determine the maximum of two integers a and b .
- *Max_3_integers_1*: The first algorithm that determines the maximum of three integers a , b , and c .
- *Max_3_integers_2*: The second algorithm that determines the maximum of three integers a , b , and c .

```

Algorithm Max_2_integers ;
Var a,b,Max : integer ;
Begin
  read(a,b) ;
  Max  $\leftarrow$  a ;
  if b > a then
    Max  $\leftarrow$  b ;
  Endif ;
  write('maximum=',Max) ;
End.

```

```

Algorithm Max_3_integers_1 ;
Var a,b,c,Max : integer
Begin
  read(a,b,c) ;
  Max  $\leftarrow$  a ;
  if b > a then
    Max  $\leftarrow$  b ;
  Endif ;
  if c > a then
    Max  $\leftarrow$  c ;
  Endif ;
  write('maximum=',Max) ;
End.

```

```

Algorithm Max_3_integers_2 ;
Var a,b,c,Max : integer ;
Begin
  read(a,b,c) ;
  if b > a then
    if b > c then
      Max  $\leftarrow$  b ;
    else
      Max  $\leftarrow$  c ;
    Endif ;
  else
    Max  $\leftarrow$  a ;
  Endif ;
  write('maximum=',Max) ;
End.

```

Questions:

- 1- Trace the algorithm **Max_2_integers** in the following cases:
 - $a = 10$ and $b = 8$
 - $a = 7$ and $b = 9$
 - $a = 6$ and $b = 6$
- 2- Based on the results from the trace in the previous question (3 cases), can we say that the algorithm is general? Justify your answer.
- 3- Are the algorithms **Max_3_integers_1** and **Max_3_integers_2** general? Provide a counter-example if the algorithm is not general.
- 4- If any of the algorithm(s) are not general, correct them.